VETERINARY OPHTHALMIC IMAGE COMPUTING

Canine Glaucoma Detection

Evaluating convolutional backbones and hierarchical Vision Transformers for automated segmentation of early canine glaucoma and preservation of vision.

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For Wisconsin Reading Center

The Need for Early Detection

What is glaucoma?

Glaucoma is caused by increased pressure inside the eye. This increased pressure can rapidly cause blindness. Glaucoma in animals is caused by the fluid inside the eye not being able to drain out effectively.

> “Canine glaucoma is a silent ophthalmic emergency. Without timely intervention, intraocular pressure spikes cause irreversible retinal ganglion cell necrosis within 24 to 48 hours”.
>
> — Veterinary Ophthalmology Clinical Guidelines & Diagnostics

Case Study: Miss Reveille X

Texas A&M's mascot, Miss Reveille X, required an eye removal due to severe, late-stage glaucoma. Because purebreds often mask early discomfort, diagnoses are frequently delayed until critical. Point-of-care computer vision models could enable routine screening, identifying lesions early enough to prevent enucleation.

Proposed Clinical Diagnostics Workflow

1

Data Ingestion & Splitting

Ingestion of canine ocular images & masks. Structured split using a 70% training, 15% validation, and 15% testing ratio.

2

Preprocessing & Augmentation

Horizontal flips, ±15° rotation, and affine translations paired with standard ImageNet normalization.

3

Architecture & Loss

Standard U-Net decoder benchmarking ResNet-34, MiT-B0, and EfficientNet-B0 backbones under joint Cross-Entropy & Soft Dice loss.

4

Model Training & Validation

Iterative optimization via AdamW and cosine annealing schedule, tracked closely with rigorous mIoU validation checkpointing.

5

Evaluation & Decision

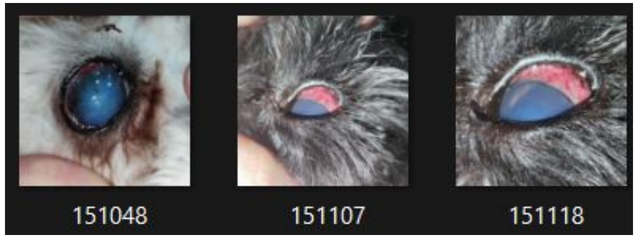
Rank models by test mIoU, load the top checkpoint for holdout inference, and flag glaucoma whenever pathological tissue is segmented.

DogEyeSeg4 Dataset Structure

Overview & Origin

Collected during routine clinical assessments in 2024 by veterinary ophthalmology specialists.

- **145 total images**: Randomized and fully anonymized.
- **Resolution**: 320x320px PNG format.
- **Purpose**: 4-Class Ophthalmic Disease segmentation to train hybrid Vision Transformer stems.



Disease Classes

- **Background:** RGB: (0, 38, 255)
- **Corneal Edema:** Gray: 1 | RGB: (0, 148, 255)
- **Episcleral Congestion:** Gray: 2 | RGB: (76, 255, 0)
- **Epiphora:** Gray: 3 | RGB: (255, 106, 0)
- **Cherry Eye Disease:** Gray: 4 | RGB: (255, 0, 110)

Encoder Comparison & Architectural Rationale

Isolating feature extraction dynamics by fixing a standard U-Net decoder across CNN and Transformer backbones.

ResNet-34

Top Performer (Highest Accuracy)

Standard convolutional network with 34 layers.

- Convolutions naturally pick up sharp edges and tiny blood vessels around the eye.
- Does not overfit easily, making it reliable on a smaller dataset of 145 images.
- Standard ImageNet pretraining delivers robust feature representation under variable illumination.

SegFormer MiT-B0

Hierarchical Vision Transformer

Lightweight Vision Transformer designed for image segmentation.

- Uses self-attention to see the big picture and connect distant areas across the image.
- Has only ~3.4M parameters, keeping the model fast and compact.
- Transformers usually need massive datasets, so on 145 images it slightly blurs small lesion borders.

EfficientNet-B0

Compound-Scaled Edge CNN

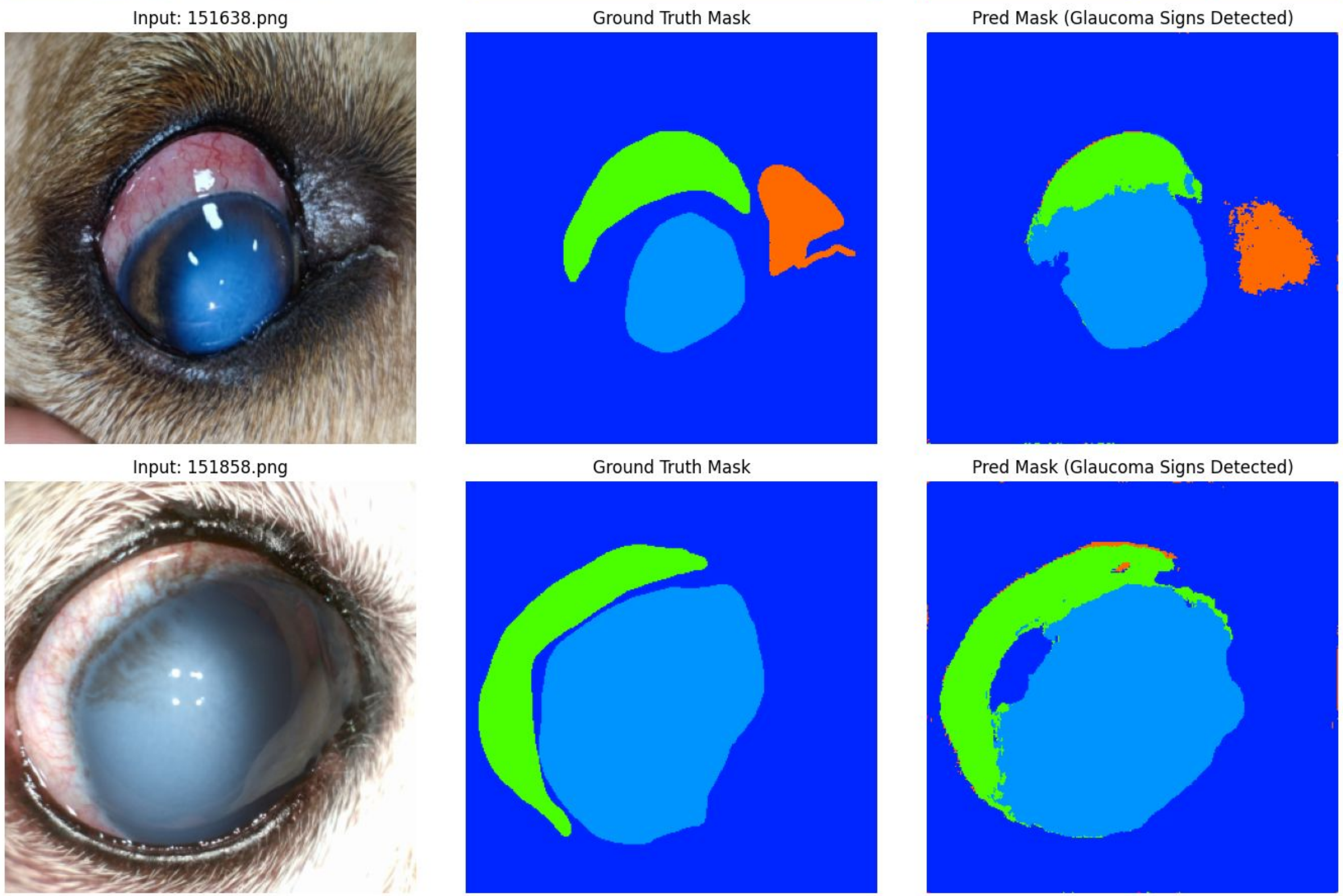
Compact convolutional network built for speed and mobile deployment.

- Automatically highlights the most useful color channels for detecting eye diseases.
- Smallest memory footprint, making it ideal for quick point-of-care screening.
- Because it compresses image details aggressively to save memory, it misses some subtle lesion outlines.

Quantitative Performance

Model	Best Val mIoU	Test Loss	Test mIoU	Test Dice (F1)	Test Precision	Test Recall	Test Specificity	Glaucoma ROC-AUC
CNN (ResNet34)	0.4278	0.6057	0.4921	0.5776	0.5954	0.5692	0.9424	**0.9524**
ViT (MiT-B0)	0.3938	0.5592	0.4617	0.518	0.5107	0.5321	0.9343	**0.8571**
CNN (EfficientNet-B0)	0.3977	0.7131	0.4463	0.5065	0.509	0.5247	0.9301	**1.0**

Model Predictions



System Deployment Roadmap

Phase 1: Architecture & Benchmarking

Evaluated ResNet-34, MiT-B0, and EfficientNet-B0 encoders with a standardized U-Net decoder.

ResNet-34 selected as top performer based on test mIoU and Dice scores.

Phase 2: Accelerated Training

GPU-accelerated training pipeline implemented in PyTorch utilizing CUDA.

Integrated Mixed Precision (AMP) for memory efficiency and faster convergence.

Phase 3: API Prototyping & Validation

Built RESTful inference endpoints using FastAPI.

Verified real-time segmentation and triage logic via interactive Swagger docs.

Phase 4: Containerization & Deployment

Built a lightweight, reproducible container using multi-stage Docker and Gradle to isolate build dependencies and deploy the FastAPI service for clinical testing.

API Prototyping

POST `/predict` Predict Eye

Parameters

No parameters

Request body required

file * required
`string`

Choose File 151556.png

Execute

151556

Response body

```
{
  "glaucoma_risk": "HIGH_SUSPECT",
  "primary_indicators": {
    "corneal_edema_detected": true,
    "episcleral_congestion_detected": true
  },
```

POST `/predict` Predict Eye

Parameters

No parameters

Request body required

file * required
`string`

Choose File 152117.png

Execute

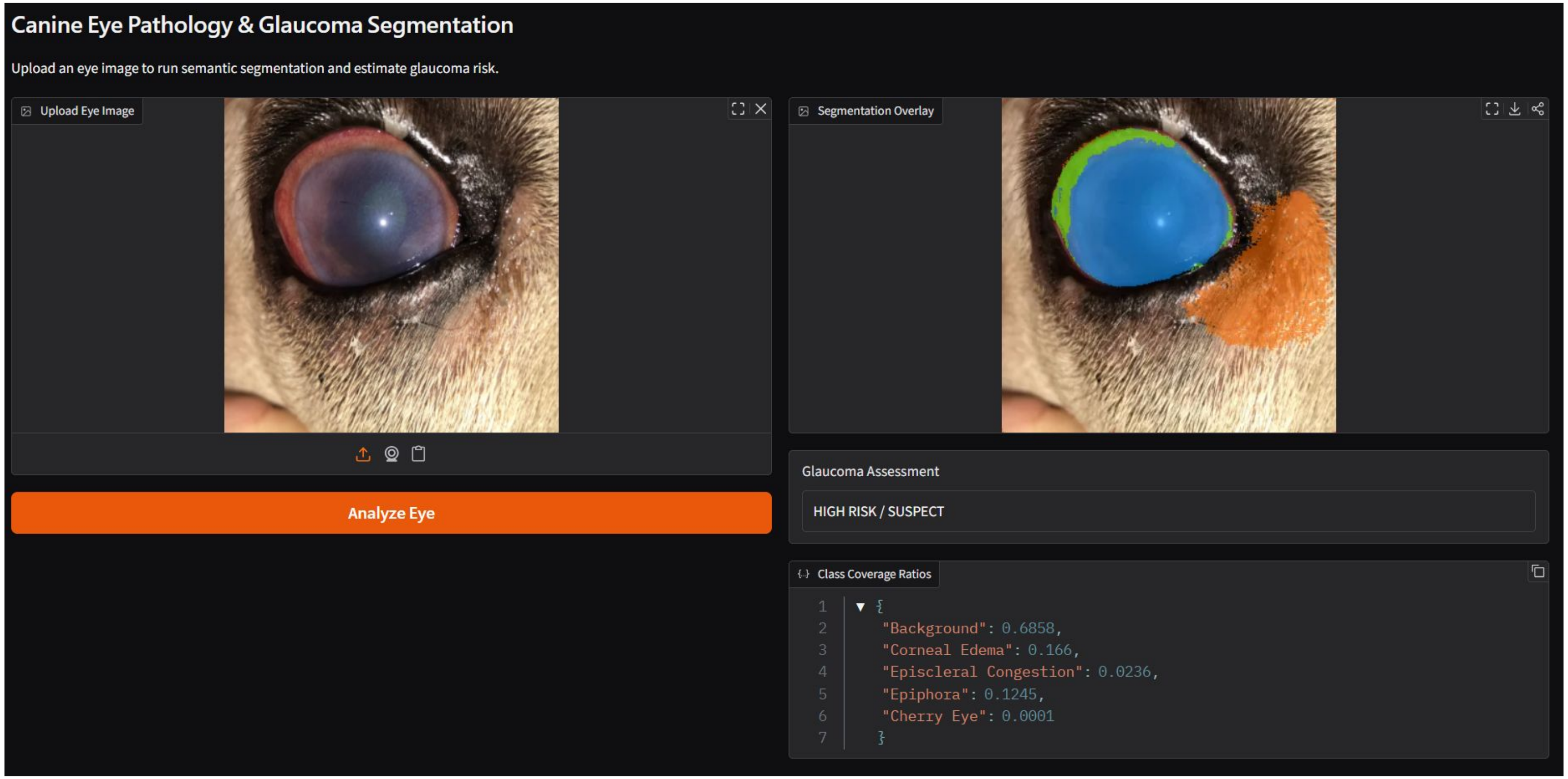
152117

Response body

```
{
  "glaucoma_risk": "LOW_NORMAL",
  "primary_indicators": {
    "corneal_edema_detected": false,
    "episcleral_congestion_detected": false
  },
```

Live Demo

https://c533589d3483ade480.gradio.live



Questions & Discussion

References:

https://www.eyevetclinic.co.uk/app/uploads/2023/04/Glaucoma-Factsheet.pdf
https://pmc.ncbi.nlm.nih.gov/articles/PMC4862370/

Resources:

Dataset: https://repository.inf.uniri.hr/en/object/infri:1309
GitHub Repo: https://github.com/AfreenAhmed405/K9_Optics